

5.2.3 Average Value

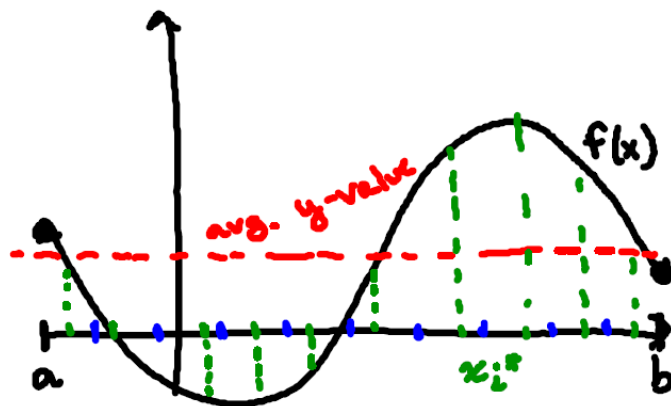
Motivation: How do you compute the average of four numbers: $a, b, c,$ & d ?

This is simply done by calculating $\frac{a+b+c+d}{4}$.

Now suppose that we wish to compute the average y -value of a function. Using the previous idea to compute such an average is much more difficult, because most functions of interest have infinitely many y -values. So, how would approach such a question?

Suppose we have a continuous function $y = f(x)$ defined on the interval $[a, b]$. We can begin by dividing up the interval $[a, b]$ into n equal subintervals each with length $\Delta x = \frac{b-a}{n}$. If we choose sample points x_i^* each from the i th interval, then we can approximate the average y -value by calculating:

$$\frac{f(x_1^*) + f(x_2^*) + \dots + f(x_n^*)}{n}$$



We should note that since $\Delta x = \frac{b-a}{n}$ then we can say that $n = \frac{b-a}{\Delta x}$ and if we use this in above calculation we have:

$$\begin{aligned} \frac{f(x_1^*) + f(x_2^*) + \dots + f(x_n^*)}{n} &= \frac{f(x_1^*) + f(x_2^*) + \dots + f(x_n^*)}{\left(\frac{b-a}{\Delta x}\right)} = \frac{[f(x_1^*) + f(x_2^*) + \dots + f(x_n^*)] \Delta x}{b-a} \\ &= \frac{1}{b-a} [f(x_1^*) \Delta x + f(x_2^*) \Delta x + \dots + f(x_n^*) \Delta x] \\ &= \frac{1}{b-a} \cdot \sum_{i=1}^n f(x_i^*) \Delta x \end{aligned}$$

Therefore, we would expect that the exact average value ought to be:

$$\lim_{n \rightarrow \infty} \frac{1}{b-a} \cdot \sum_{i=1}^n f(x_i^*) \Delta x = \frac{1}{b-a} \cdot \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

We recognize this as a definite integral!!!

Definition: The **average value** of f on $[a, b]$ is:

$$f_{avg} = \frac{1}{b-a} \int_a^b f(x) dx$$

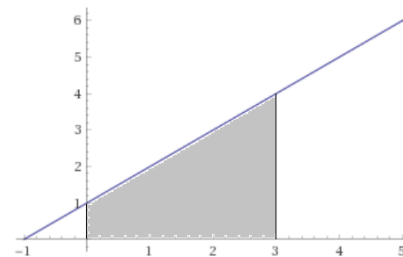
Example 1: Find the average value of the function $f(x) = 1 + x$ on the interval $[0, 3]$.

$$f_{avg} = \frac{1}{3-0} \int_0^3 (1+x) dx = \frac{1}{3} \cdot \int_0^3 (1+x) dx$$

We can use geometry to find the area under $y = 1 + x$ on the interval $[0, 3]$.

$$A = \int_0^3 (1+x) dx = 3(1) + \frac{1}{2}(3)(3) = 3 + \frac{9}{2} = \frac{15}{2}$$

$$\text{Therefore we have } f_{avg} = \frac{1}{3} \cdot \int_0^3 (1+x) dx = \frac{1}{3} \cdot \left(\frac{15}{2} \right) = \frac{5}{2}.$$



Example 2: Find the average value of the function $f(x) = \sqrt{1-x^2}$ on the interval $[-1, 1]$.

$$f_{avg} = \frac{1}{1-(-1)} \int_{-1}^1 \sqrt{1-x^2} dx = \frac{1}{2} \cdot \int_{-1}^1 \sqrt{1-x^2} dx$$

We can use geometry to find the area under $y = \sqrt{1-x^2}$ on the interval $[-1, 1]$.

$$A = \int_{-1}^1 \sqrt{1-x^2} dx = \frac{1}{2} \pi (1)^2 = \frac{\pi}{2}$$

$$\text{Therefore we have } f_{avg} = \frac{1}{2} \cdot \int_{-1}^1 \sqrt{1-x^2} dx = \frac{1}{2} \cdot \left(\frac{\pi}{2} \right) = \frac{\pi}{4}.$$



Question: If we are always finding the average value of a continuous function, must it be the case that there is a value of x where $f(x) = f_{avg}$?

Intuitively this seems true. If the function is continuous it would have to pass through (at some point) the y -value that is the calculated to be f_{avg} .

The next theorem makes this idea precise.

Theorem (The Mean Value Theorem for Integrals): If f is continuous on $[a, b]$, then there exists a number c in $[a, b]$ such that:

$$f(c) = f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx$$

Proof:

Since f is given as a continuous function on $[a, b]$, then by the Extreme Value Theorem f has an absolute maximum value M and an absolute minimum value m on the interval $[a, b]$.

Therefore we can say that $m \leq f(x) \leq M$.

This immediately implies that $\int_a^b m dx \leq \int_a^b f(x) dx \leq \int_a^b M dx$.

We calculate the outer integrals and obtain: $m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$.

If we divide everything by $(b-a)$ we obtain: $m \leq \frac{1}{(b-a)} \int_a^b f(x) dx \leq M$.

But then, by the IVT, $f(x)$ must equal $\frac{1}{b-a} \int_a^b f(x) dx$ for some value c in $[a, b]$!

Therefore we can say that for some c in $[a, b]$ we have $f(c) = f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx$.

■

Example 3: Find all numbers c that satisfy the MVT for Integrals for the function $f(x) = 1+x$ on the interval $[0, 3]$.

Since $f(x) = 1+x$ is continuous on $[0, 3]$, the MVT for integrals states that there is a c in $[0, 3]$ such that:

$$f(c) = \frac{1}{3-0} \int_0^3 (1+x) dx = \frac{5}{2}$$

So we have $1+c = \frac{5}{2} \Rightarrow c = \frac{3}{2}$.

Example 4: Find all numbers c that satisfy the MVT for Integrals for the function $f(x) = \sqrt{1-x^2}$ on the interval $[-1,1]$.

Since $f(x) = \sqrt{1-x^2}$ is continuous on $[-1,1]$, the MVT for integrals states that there is a c in $[-1,1]$ such that:

$$f(c) = \frac{1}{2} \cdot \int_{-1}^1 \sqrt{1-x^2} dx = \frac{\pi}{4}$$

$$\text{So we have } \sqrt{1-c^2} = \frac{\pi}{4} \Rightarrow 1-c^2 = \left(\frac{\pi}{4}\right)^2 \Rightarrow c^2 = 1 - \left(\frac{\pi}{4}\right)^2 \Rightarrow c = \pm \sqrt{1 - \left(\frac{\pi}{4}\right)^2}.$$

Both of these values are in the interval $[-1,1]$ and so both of them are acceptable values of c .

Example 5: If f is a continuous function and $\int_2^7 f(x) dx = \frac{11}{2}$, show that f takes on the value of 1.1 at least once on the interval $[2,7]$.

Since f is continuous then by the MVT for integrals, there exists a c in $[2,7]$ such that:

$$f(c) = \frac{1}{7-2} \cdot \int_2^7 f(x) dx = \frac{1}{5} \cdot \frac{11}{2} = \frac{11}{10} = 1.1$$

Note: The Average Value of a function and the MVT for Integrals are powerful concepts that will help us to connect the various ideas of Calculus in what we call the Fundamental Theorem of Calculus!!!